Coimisiún na Scrúduithe Stáit
State Examinations Commission

Junior Cycle Examination 2027

Applied Technology

Coursework Briefs

Common Level

To be completed by 30 April 2027

220 marks

Instructions to candidates

1. Select **one** brief from the three options provided.
2. Your completed coursework submitted for assessment must consist of **two** items:
    - an **Artefact** *and*
    - a **My Design Journey** folio.
3. Please familiarise yourself with the adjusted assessment arrangements for candidates taking the state examinations in the 2026/27 school year, that were issued by the Department of Education and Youth. As a result of these adjustments only three of the six headings in your My Design Journey folio must be presented in 2027. The other three headings are not examinable and will not have marks allocated to them.
   Therefore, candidates are only required to complete the following **three** headings in 2027:
    - **My Preferred Design** & **justification**
    - **Manufacturing sequence**
    - **Evaluation** & **Reflection on learning.**

   As a result of these changes, the total mark allocation for this coursework in 2027 is 220 marks, instead of the usual 280 mark total. However, the relative weighting between the coursework (70%) and the written paper (30%) will be retained. This will be achieved by reweighting the coursework before it is combined with the mark for the written component.
4. Your Design Journey folio should consist of a maximum of **10** single-sided A4 pages in total. Your folio can be presented in A4, A3, or a combination of both formats. Each A3 page presented equals two A4 pages.
5. Coursework should include an appropriate range of materials, skills, processes, and techniques (a variety of manual processing skills should be demonstrated).
6. You must submit your original sketches in your folio.
7. Where specialised processes (CAD, CAM, control technology etc.) are used, these should be supported in your Design Journey folio by the inclusion of drawings/flowcharts/program code and/or descriptions/photographs as appropriate.
8. Electronic/electrical circuitry (such as circuit boards and controls) should be neatly housed and accessible for examination purposes.
9. Electrical power sources should not exceed 12V DC.
10. Primary research should be included where appropriate.
11. When using research sources, including the Internet and material generated using artificial intelligence (AI) software or applications, the sources must be acknowledged. Material copied directly from the Internet or from other sources and presented as your own work will not receive any marks.
12. Your **Examination Number** must be written clearly on both your completed artefact and on your Design Journey folio.
13. The completed coursework, artefact and folio, must be available for assessment by **30 April 2027**.

Authentication of coursework:

1. The coursework submitted for assessment must be **your own individual work** and must be completed in school under the supervision of the class teacher.

2. Your artefact must not be removed from the school under any circumstances as doing so may result in the coursework being considered invalid.

3. It is your responsibility, as the candidate, to comply with these requirements. If you fail to comply with these requirements, your teacher will not be able to validate, and sign off, on your coursework and you will not receive any marks for it.

4. Coursework must be completed in full and strict compliance with the procedures and conditions outlined in this coursework brief and in the State Examinations Commission (SEC) Coursework Rules and Procedures.

Note:

- *The circumstances in which the Minister for Education and Youth may withhold marks from candidates are set out in the "Rules and Programme for Secondary Schools".*
- *Anyone who helps a candidate to break examination rules can be prosecuted under the Education Act 1998.*

Coursework Briefs 2027

Select **one** of the following briefs.

1. **Behind every breath-taking moment on a modern theatre stage lies a hidden world of electro-mechanical prop design. Stage props are the essential, portable objects that actors interact with to breathe life and reality into a live performance.**

   In the context of a particular theatrical show, design and make an electro-mechanically operated prop that can be reused for successive performances of that show.

   Your solution should incorporate a control system of your choice and be aesthetically pleasing.

   Note: Your solution may be a scaled model or full size in the context of a smaller artefact.

2. **Animal welfare focuses on ensuring that animals under human care live healthy, comfortable, and stress-free lives by providing proper nutrition, safe environments, and opportunities to express their natural behaviours. Animal hygiene, feeding systems, environmental controls and measures that enhance physical and psychological welfare are essential parts of animal care and husbandry.**

   In the context of an animal or animals of your choice under human care, design and make an electro-mechanical device/system that caters to an essential need of your selected animal(s).

   Your solution should incorporate a control system of your choice and be aesthetically pleasing.

   Note: Your solution may be a scaled model or full size in the context of a smaller artefact.

3. **Charitable organisations regularly devise different ways to raise money from the public. They often rely on public coin collection devices as a simple and accessible way for individuals to donate their spare change.**

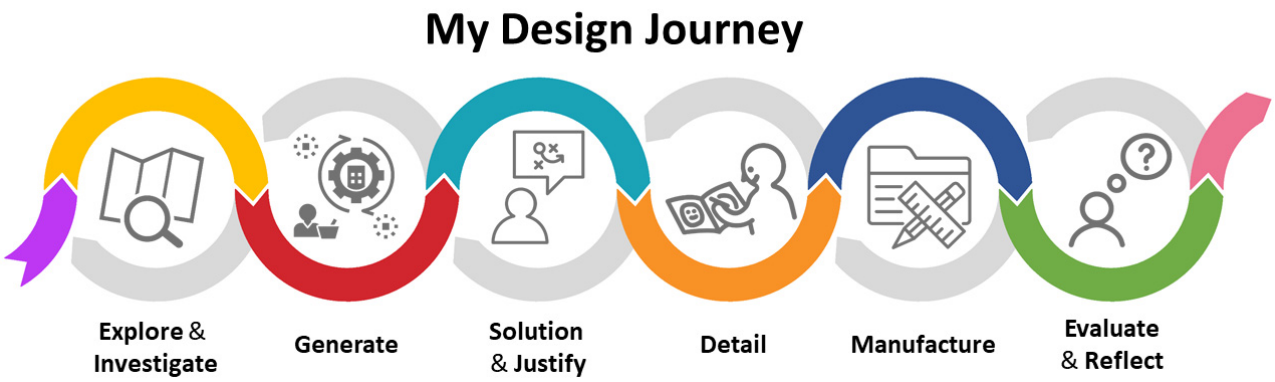
   In the context of a charity of your choice design and make a fun, interactive coin collection device with a suitable electro-mechanical feature. This collection device should be designed for the countertop of a retail outlet, or as a wall fixture in such an outlet.

   Your solution should incorporate a control system of your choice and be aesthetically pleasing.

Information for candidates

Your Design Journey

The graphic below shows a design process which you could follow in the design and realisation of your Applied Technology coursework. This graphic maps your journey through a sequence of engagements with the coursework folio and artefact. On completion of this journey, you should have a completed folio and artefact and will have gained knowledge, skills and experience through undertaking the process.
As outlined earlier, only three headings of the design journey folio are examinable in 2027.



Allocation of Marks

The completed coursework will be assessed using the following mark allocations:

- My Design Journey folio 60 marks
- Artefact 160 marks

Note: For 2027, the total marks for the coursework will be 220 marks instead of 280 marks. The 70% coursework weighting (280 marks) will be retained by re-weighting the coursework mark before it is added to the written mark.

Presentation of Coursework

On completion of the coursework, the class teacher should ensure that the finished Artefacts and corresponding Design Journey folios are presented in numerical order for assessment as listed on the P2 form.

Storage of Coursework

The finished Artefact and Design Journey folio are kept in a secure place until the examining process (including appeals) has concluded.

Outline Marking Scheme

An outline marking scheme that will be used to assess your completed coursework is given below. While the general mark allocations will largely remain the same, assumptions about future mark breakdowns should be avoided.

My Design Journey - Folio			
Phases of Design Journey	**Description**	**Suggested pages**	**Marks**
Phase 1 **Exploration and Investigation of Brief**	***Not examinable for 2027***		
Phase 2 **Generation of Ideas**	***Not examinable for 2027***		
Phase 3 **My Preferred Solution** *and* **Justification**	Presentation of your preferred design solution. Outline and justify the design decisions made to arrive at your preferred solution (to include a justification of the selected mechanical and electrical/electronic sub-systems).	2 – 3	20
Phase 4 **Details for Manufacture**	***Not examinable for 2027***		
Phase 5 **Manufacturing Sequence**	Detailed record of all manufacturing processes (include electronic/electrical and mechanical elements) and the sequence used to complete the artefact (include photos where appropriate). Details of any modifications made during production and reasons for such changes (if and where relevant).	3 – 4	25
Phase 6 **Evaluation** and **Reflection on learning**	Evaluate and reflect on your learning through completing your coursework using these headings: **Successes**, **Challenges**, **Skills** and **Other Learning Experiences**.	1 - 3	15
Artefact			
Artefact satisfies brief	The requirements of the chosen brief are fulfilled by the artefact presented.		20
Creativity and Inventiveness	Expression of individual creativity and inventiveness in the **unique** solution developed and the artefact presented.		20
Functionality	Do all elements of the artefact function as planned in the Design Journey folio?		20
Appropriate use of materials, processes, components, and techniques	Materials, processes, components and techniques used which are appropriate to their purpose and function. Appropriate use of commercial components.		20
Production skills and processes	Processing of materials showing a range and depth of appropriate and varied skills (including manual processing skills and skills associated with the electronic/electrical system) with due regard to health and safety.		35
Assembly	How well the artefact is assembled using appropriate joining methods.		25
Detailed finish and overall presentation	High quality finishing, well presented with all parts well integrated and operating controls labelled where appropriate.		20
		Overall Coursework Total in 2027	**220**